

ABINAYA K

9944818490 | abinayak829@gmail.com | LinkedIn | GitHub | LeetCode (250+) | GeeksforGeeks (150+) | Portfolio

EDUCATION

Higher Secondary Course

Dr. V.G.N. Matric Higher Secondary School, Ranipet.

Percentage: 98.5

Sep. 2021 – Apr. 2023

B.Tech Artificial Intelligence and Data Science

Madras Institute of Technology, Anna University, Chennai.

CGPA: 8.88

Aug. 2023 – Present

PROJECTS

FinGuard AI: Explainable Financial Fraud Detection using Graph Intelligence and LLMs (Ongoing)

Mar. 2026 – Present [\[GitHub\]](#)

- Designing a **graph-based fraud detection system** to identify anomalous patterns such as fraud rings and abnormal transaction flows.
- Developing a **real-time pipeline using Python and REST APIs** for efficient risk scoring of transactions.
- Integrating **LLM-based reasoning** for human-readable fraud explanations while exploring optimization techniques for scalable graph analysis.

Diet Monitoring System – Flask (Python), SVM, Computer Vision

Feb. 2025 – Apr. 2025 [\[GitHub\]](#)

- Built an end-to-end **food analysis system** using image segmentation and **SVM-based classification**, achieving **78% accuracy**.
- Implemented **calorie prediction** using visual features and nutritional datasets.
- Developed a **Flask-based backend** to process user-uploaded images and generate real-time dietary insights.

Digital Sanitation Tracker – React, Node.js, PostgreSQL, Socket.IO

Aug. 2025 – Sept. 2025 [\[GitHub\]](#)

- Designed a **real-time sanitation management system** with **event-driven communication** using Socket.IO.
- Built scalable **REST APIs** with Node.js and PostgreSQL for structured data management.
- Implemented **role-based task assignment** to improve workflow efficiency and issue resolution.

Remaining Useful Life (RUL) Prediction using TCN with Attention

Jan. 2026 – Mar. 2026 [\[GitHub\]](#)

- Built an end-to-end **RUL prediction system** using deep learning on NASA CMAPSS time-series data with optimized preprocessing.
- Developed a **TCN with attention** for temporal modeling and deployed it via a **Flask REST API** for real-time inference. Optimized performance and achieved **~13 RMSE** on unseen test data, while exploring further model improvements.

VIRTUAL INTERNSHIP

Infosys Springboard Virtual Internship (Remote)

Feb. 2026 – Apr. 2026 [\[GitHub\]](#)

- Completed a virtual internship focused on **parallel text processing** using Python. Built a Python-based system for handling large-scale text processing tasks. Applied optimization techniques to improve processing performance.

SKILLS

- **Programming Languages:** Python, C, C++, Java (Basics)
- **Core Skills:** Data Structures and Algorithms (250+ Leetcode problems), System Design Fundamentals
- **Technologies:** REST APIs, MySQL, OracleDB, Git, Firebase, Cloud (Basics)
- **Relevant Courses:** Design Thinking – A Primer (NPTEL), Artificial Intelligence Fundamentals (IBM), Data Structures and Algorithms (AU-MIT), Database Management Systems (AU-MIT), Operating Systems (AU-MIT).
- **Research Interests:** Explainable AI, Graph Machine Learning, Computer Vision, Quantum Machine Learning

ACTIVITIES

- **NSS Volunteer:** Contributed to social service projects through the National Service Scheme.
- **QUANTuMIT Club Member (President of Technical Activities):** Explored quantum mechanics and computing through interactive sessions and workshops.
- **Member of Tamil Mandram (Student club of MIT):** Participated as a speaker in Tamil Mandram podcasts.

HACKATHONS

Fish Recognition with Chatbot Integration (Agentic Hackathon Finalist)

- Reached the finals of the Agentic Hackathon, showcasing the project's innovation in merging AI-driven vision and conversational AI.